HACT PS3

Judith Kleman & Félix Ranson

due November 5th 2025

1 Solving Krusell-Smith in continuous time

1.1 Defining the RCE

A recursive competitive equilibrium of the Krussel-Smith economy in continuous-time is a collection of function $v(a, y, z, K)$, $c(a, y, z, K)$, $s(a, y, z, K)$, $r(z, K)$, $w(z, K)$, $g(a, y, t)$, and $h(K, z)$ such that:

- v, c, s solve the household optimization problem[1] *i.e.*the HJB equation given by

$$\rho v(a, y; z, K) = max_c u(c) + \mathcal{A}_c v + \mathcal{Q}_v + \mathcal{L}_v + \mathcal{A}_k v$$

  with $\mathcal{A}_c$ the generator of the earnings process, Λ the generator of the earnings process, $\mathcal{L}$ the generator of the TFP process, and $\mathcal{A}_K = \frac{\partial v}{\partial K} h(K, z)$ the generator of the perceived law-of-motion of aggregate capital. This latter part captures how households summarize the evolution of the entire joint distribution g through a finite set of moments.

- Firms maximize instantaneous profits under competitive markets: r and w are set competitively *i.e.* $r_t = \alpha e^{z_t} \left(\frac{K_t}{\bar{L}}\right)^{\alpha-1} - \delta$ and $w_t = (1-\alpha) e^{z_t} \left(\frac{K_t}{\bar{L}}\right)^{\alpha}$

- Markets clear at each time t:

$$K_t = \sum_y \int a \; g(a, y, t) \; da \text{ and } L_t = \sum_y \int y \; g(a, y, t) \; da$$

  where $g(a, t, t)$ is given by the Kolmogorov Forward equation:

$$\partial_t g_t(a, y) = -\partial_a [s(a, y, r_t, w_t) g_t(a, y)] + \lambda(y_{tilde}) g_t(a, y) - \lambda(y) g_t(a, y)$$

  The aggregate capital and labor are then given by market clearing:

- The actual path of aggregate capital K_t and the perceived law of motion are consistent:

$$dK_t = h(K_t, z_t) dt$$

1.2 Principle of the solution

We discretize the state-space over four grids $\mathbb{A}, \mathbb{Y}, \mathbb{Z}, \mathbb{K}$. The principle of the algorithm is as follows

- First we simulate a time path of length T of the aggregate shock by Euler-Maruyama discretization : $z_{t+\Delta} = -\eta(\bar{z} - z_t)\Delta + z_t + \sigma\varepsilon\sqrt{\Delta t}$ where $\varepsilon \sim \mathcal{N}(0, 1)$

- We then choose the number of moments m of the distribution of the aggregate endogenous state variable K that households try to forecast. We set $m = 1$

- We choose a functional form for $dK_t \equiv h(K_t, z_t)dt$. We take $h(K_t, z_t) = K_t(\theta_0 + \theta_1 \log(K_t) + \theta_2 z_t + \theta_3 \log(K_t) z_t)$

[1] The household solves:

$$\begin{aligned} v(a, y, z, g) = \sup_{(c_t)_{t \geq 0}} \mathbb{E}\left[\int_0^\infty e^{-\rho t} \, u(c_t) \, dt\right] \\ \text{s.t. } da_t =& (r(g_t, z_t) a_t + w(g_t, z_t) y_t - c_t) dt \\ dz_t =& -\eta(\bar{z} - z_t) + \sigma dW_t \\ dy_t =& (\tilde{y} - y_t) dN_t \end{aligned}$$

- Conditional on a vector of parameters Θ, we solve the household's problem on our the grids $\mathbb{A}, \mathbb{Y}, \mathbb{Z}, \mathbb{K}$
- We solve for the law of motion of the distribution of households' holdings $g(a, y)$ through the Kolmogorov Forward Equation and ensure market clearing at each stage by interpolation.
- We estimate new parameters for h based on the actual path of aggregate capital induced by the law of motion of the distribution. If the new parameters are close enough to the previous ones we break an measure the model's goodness of fit. If not, we update Θ through a relaxation scheme.

Solving the HJB equation of the households requires us to discretize four infinitesimal generators : $\mathcal{A}_c$ for the wealth process, Λ for the earnings process, $\mathcal{L}$ for z and $\mathcal{A}_k$ for the perceived law-of-motion of the aggregate states. Λ is simply given by $\begin{pmatrix} -\lambda_1 & \lambda_1 \\ \lambda_2 & -\lambda_2 \end{pmatrix}$ and $\mathcal{L}$ by the usual OU-process generator (*c.f.* Problem Set 2). $\mathcal{A}_k$is simply the upwind finite difference matrix of the perceived law of motion where the drift is given by $h(K, z)$. To build $\mathcal{A}_c$, we first need to build the drift of wealth from $\dot{a} = w(z, K)y + r(z, K)a - u'^{-1}(v_a(a, y, z, K))$ given a choice of v. This is done by upwind finite differences on the asset grid, as usual. The important change compared to the Aiyagari model is that v is defined over the grids $\mathbb{A}$, $\mathbb{Y}$, $\mathbb{Z}$ and $\mathbb{K}$, so that for each pair of aggregate states (z, K), we have to compute the drift of wealth for all income states y. There is therefore one blockdiagonal matrix $A_c^{iy,iz,ik}$ for each idiosyncratic income state y_{iy}, and each aggregate state pair (z_{iz}, K_{ik}). We stack them into the $(Na * Ny * Nz * Nk)^2$ blockdiagonal matrix A_c. To build the full generator of the HJB, we then recast all generators as $(Na * Ny * Nz * Nk)^2$ matrices and sum :

$$A = A_c + (I_{Nk} \otimes I_{Nz} \otimes \Lambda \otimes I_{Na}) + (I_{Nk} \otimes L \otimes I_{Ny} \otimes I_{Na}) + (A_K I_{Ny} \otimes I_{Na})$$

And finally we solve the HJB using the implicit method:

$$v^{n+1} = \left(\left(\rho + \frac{1}{\Delta}\right) I - A\right)^{-1} (u^n + \frac{1}{\Delta} v^n)$$

This step is repeated until v converges.

After solving the HJB, we want to simulate how the distribution of agents evolves over time, and thus the aggregate capital stock. Starting from an initial distribution $g_0(a, y)$ - and thus an intial capital stock K_0 we solve the Kolmogorov Forward Equation (KFE). The dependance on the aggregate state manifests itself through the right choice of the block of the generator of the wealth process corresponding to the aggregate state (z_t, K_t) at each time t. However, during the simulation, the aggregate state (z_t, K_t) continuously varies over time and will not exactly coincide with the grid points. Therefore, at each time step t, we must interpolate between the four neighboring corner operators (z^d, K^d), (z^d, K^u), (z^u, K^d), (z^u, K^u). After searching for the position of z_t and K_t [2], we can compute the 4 corner operators

$$A_{dd} = A_c^{(z_d, K_d)} + \Lambda \otimes I_{Na}, \qquad A_{du} = A^{(z_d, K_u)} + \Lambda \otimes I_{Na}, \qquad A_{ud} = A^{(z_u, K_d)} + \Lambda \otimes I_{Na}, \qquad A_{uu} = A^{(z_u, K_u)} + \Lambda \otimes I_{Na}.$$

Then, for each operator, we compute the distribution after Δ_t with the implicit scheme

$$g_{t+\Delta t}^{corner} = (I - \Delta t A_{corner}^T)^{-1} g_t$$

which gives the distribution $g_{dd}, g_{du}, g_{ud}, g_{uu}$. We can now reconstitute the true distribution at the point (z_t, K_t), by doing a bilinear interpolation with the weight (ω_z, ω_K):

$$g_{t+\Delta_t} = \omega_z \omega_K g_{uu} + (1 - \omega_z)\omega_K g_{du} + (1 - \omega_K)\omega_z g_{ud} + +(1 - \omega_z)(1 - \omega_K) g_{dd}$$

[3]. We can then compute the aggregated capital over the period :

$$K_t = \sum_y \int a \; g_t(a, y) da$$

The rest of the algorithm is straightforward : parameters of the perceived law-of-motion are updated by OLS on the actual path of capital until convergence.

[2] $z_t = (1 - \omega_z) z_{izd} + \omega_z z_{izd}$, with izd the superior indice of z, izu the inferior, and ω_z the linear interpolation weight of z

[3] Which is also a weighted mean of the four distribution according to the real position of (z_t, K_t)

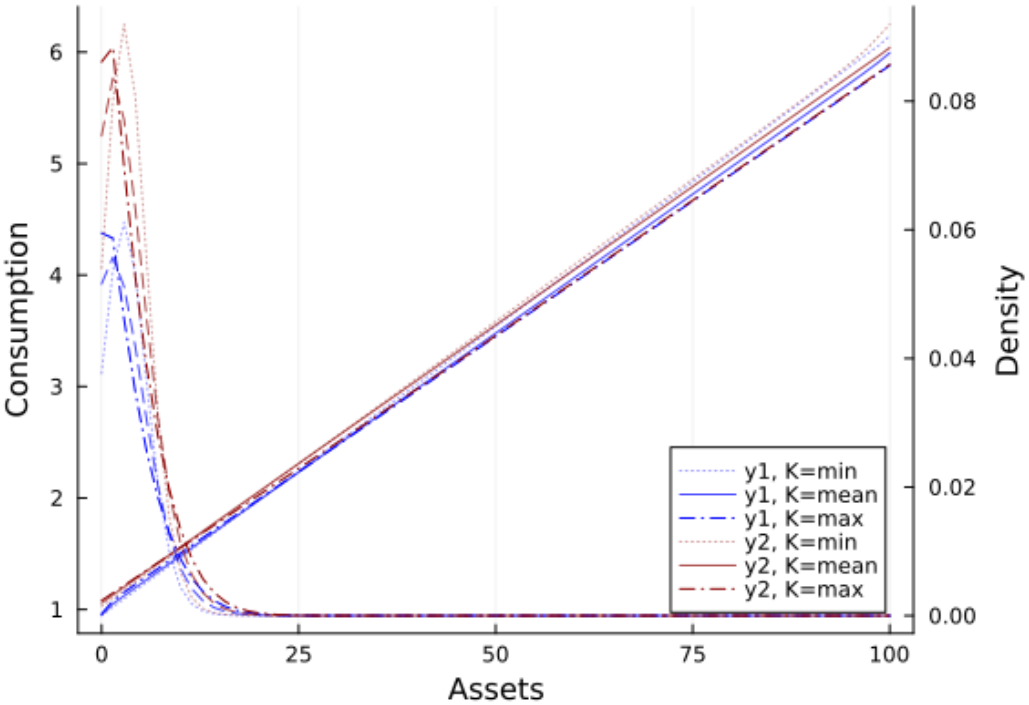

Figure 1: Consumption policy function and cross-sectional distribution for $z = \bar{z}$ and different values of K and y

1.3

We plot the cross-sectional distribution for both income states as well as the consumption policy functions. We de so for both income states y_1, y_2, $K = \{\mathbb{E}_t(K_t), \min_t K_t, \max_t K_t\}$ (which do not lie on the grid, so we interpolate these values), and $z = \bar{z}$. We can see that most of the distribution is concentrated at the bottom of the asset space with a thin tail ($\sim$ log normal). There's more people at $y = y_2$ - which makes sense since $\lambda_1 > \lambda_2$. The consumption policy function is almost exacly linear except at the very bottom of the asset space and for low values of y. Movements in K only slightly alter the policy function and the distribution. The precautionary savings motive linked to incomplete market is thus weak.

1.4

The R^2 of the final regression is 0.9736061241901125, which is very high although a bit lower than in the original KS article which is set in discrete time and uses a different TFP process. This value indicates that households are making reliable predictions and that their perceived law-of-motion accounts for most of the variance in aggregates that they indeed witness. Den Haan proposes to measure the accuracy as the maximum percentage difference between the path of capital calculated as $K_t = \sum_y \int a g_t(a, y) da$ and $K_t = K_{t-1} + h(K_{t-1}, z_{t-1})dt$. This is because the R^2 might be high even though households sometimes make potentially large mistakes that cumulate over time. Indeed, the R^2 of a regression mesure the part of the variance accounted for by explanatory variables, but if the dependant variable varies little and if the errors are small compared to the variance of K_t, the R^2 could be close to 1, even though the cumulated forecasting errors are large in levels. Figure 2 shows the two series. Overall, they match really well, but we can still see some deviations (especially downwards, households seem to anticipate bigger declines in the capital stock than those that actually occur). The Den Haan metric is equal to $0.006222085296943534 \approx 0.6\%$, which is small.

1.5

This model features approximate aggregation, but for different reasons than the usual KS model. Indeed, here the bottom of the distribution is much more populated than the rest of the state space. But policy functions are almost exactly linear, and the precautionary savings motive is very mild. This is likely because of the properties of the stochastic processes we use : η is high so that TFP reverts to the mean very quickly (persistence is low), σ is low so the distance to the mean is low too, and $\lambda_1 > \lambda_2$ which means that households exit the low income state more often than the high income state. All in all, households know that aggregate shocks quickly vanish and are not very sizeable, and they expect to spend more time in the high income state. This tends to make

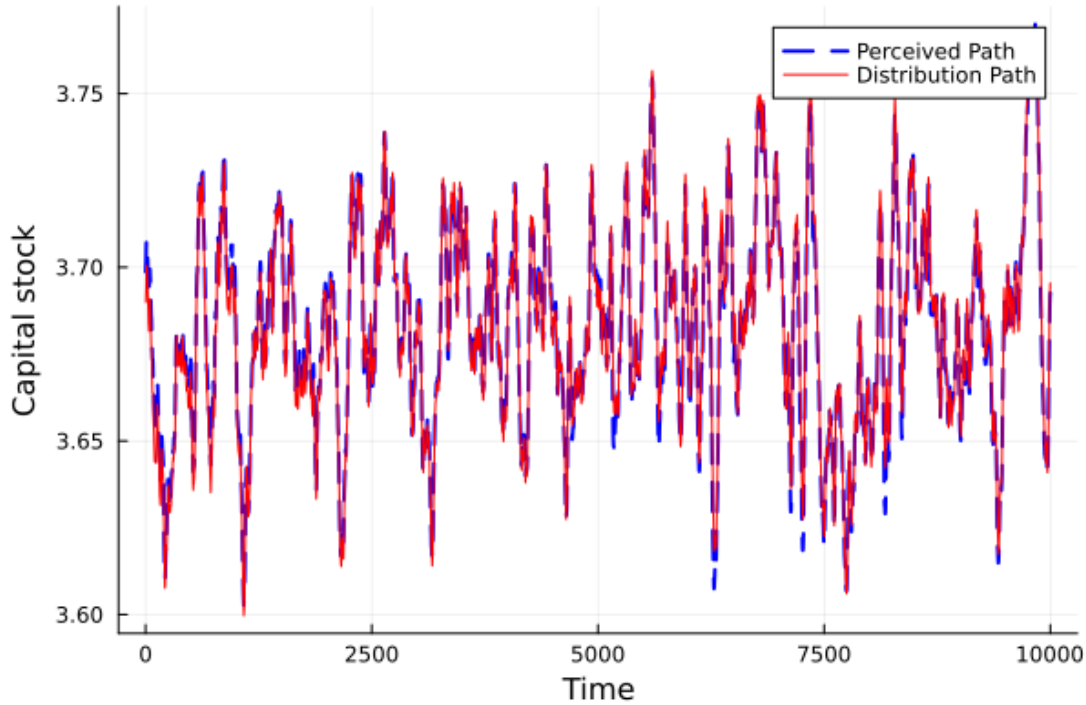

Figure 2: Comparison of aggregate capital paths

the precautionary motive less important, so that households start to smooth consumption very "early" on the asset grid.

2 Fokker-Planck Equation with common noise

Let us start from the SDE for x_t defined on the probability space $(\Omega, \mathcal{F}, \mathcal{P})$

$$dx_t = \mu_t(x_t) + \sigma_t(x_t)dW_t + \sigma_t^0 dW_t^0$$

dW_t^0 is common noise and dW_t is idiosyncratic noise such that $\int dW_t dG = 0$ where $G(x,t)$ is the c.d.f. of x at time t, conditional on some value of the shock dW_t^0. We always integrate on the support $x(\Omega)$ but to ease notations we do not always write $\int_{x \in x(\Omega)}$. We also drop the explicit dependence on t and on x for most of our calculations and re-introduce it only when it is necessary. Also I always write sup or inf even though this object may not exist and be ∞ but this just for notational compactness
Now take any test function $f \in \mathcal{C}^2(x(\Omega))$ with compact support and such that f' and f'' also have compact support. We assume that f and its derivatives vanish on the boundary of their support. By Ito's lemma, we have:

$$df(x_t)\Big|_{x_t=x} = \left(f'\mu + \frac{1}{2}(\sigma^2 + \sigma_0^2)f''\right)dt + f_x\sigma^2 dW + f_x\sigma_0^2 dW^0$$

We next take the cross-sectional expectation conditional on a value W_0 of common noise, *i.e.* we integrate with respect to the measure $dG = g(x,t)dx$:

$$\int df dG = \int \left(f'\mu + \frac{1}{2}(\sigma^2 + \sigma_0^2)f''\right)dtdG + \int f'\sigma_0^2 dW^0 dG$$

We don't want to "divide by d_t" because this wouldn't be properly defined for dW_t^0. So instead, we keep the formulation as it is and perform integration by part on the r.h.s. Let us start with $\int f\mu dG = \int f'(x)\mu_t(x)g(x,t)dx$. We have :

$$\int f'(x)\mu_t(x)g(x,t)dtdx = [f(x)\mu_t(x)g(x,t)dt]_{\inf x(\Omega)}^{\sup x(\Omega)} - \int f(x)\frac{\partial}{\partial x}(\mu_t(x)g(x,t)d)dtdx$$

The term in square brackets is equal to zero because we have assumed that f vanishes at the boundary of the support, so that we're left with only the integral term. Coming now to the second part of the r.h.s. :

$$\int (\sigma^2 + \sigma_0^2 f'' dt dG = \left[f'(x)(\sigma^2(x) + \sigma_0^2(x))g(x,t)\right]_{\inf x(\Omega)}^{\sup x(\Omega)} - \int f'(x)\frac{\partial}{\partial x}\left\{(\sigma^2(x) + \sigma_0^2(x))g(x,t)\right\} dtdx$$
$$= -\left[f(x)\frac{\partial}{\partial x}\left\{(\sigma^2(x) + \sigma_0^2(x))g(x,t)\right\} dt\right]_{\inf x(\Omega)}^{\sup x(\Omega)} + \int f(x)\frac{\partial^2}{\partial x^2}\left\{(\sigma^2(x) + \sigma_0^2(x))g(x,t)\right\} dtx$$

Where again the terms in brackets cancelled because of our assumptions on f. And for the last term:

$$\int f'\sigma_0^2 dW^0 g(x,t)dx = \left[f(x)\sigma_0^2(x)g(x,t)dW\right]_{\inf(x(\Omega))}^{\sup(x(\Omega)} - \int f(x)\frac{\partial}{\partial x}\left(\sigma_0^2(x)g(x,t)\right) dW^0 dx$$

By the same argument, we're left only with the integral term. Summing up we have:

$$\int df(x_t)\Big|_{x_t=x} g(x,t)dx = \int f(x)\left(-\partial_x(\mu g)dt + \frac{1}{2}\partial_{xx}\left[(\sigma^2 + \sigma_0^2)g\right] dt - \partial_x(\sigma_0^2 g)dW_t^0\right) dx$$

Coming now to the l.h.s we have:

$$\int df(x_t)g(x,t)dx = \mathbb{E}(df(x_t))\Big|_{x_t=x} = \lim_{\Delta t \longrightarrow 0} \mathbb{E}(df(x_{t+\Delta t})) - \mathbb{E}(df(x_t)) = \lim_{\Delta t \longrightarrow 0} \int f(x)g(x,t+\Delta t)dx - \int f(x)g(x,t)dx$$

where the last equality follows from the fact that f doesn't depend on time. Next we use the dominated convergence theorem[4] to switch limits to get:

$$\int df(x_t)g(x,t)dx = \int f(x)\left(\lim_{\Delta \longrightarrow 0} g(x,t+\Delta t) - g(x,t)\right) dx = \int f(x)dg(x,t)dx$$

Since f is a test function, for the two integrals to be equal, the integrands must be equal so that:

$$dg(x,t) = -\partial_x(\mu g)dt + \frac{1}{2}\partial_{xx}\left[(\sigma^2 + \sigma_0^2)g\right] dt - \partial_x(\sigma_0^2 g)dW_t^0$$

[4] Assuming that g is continuous in t and locally bounded in x